

Application Responses - Senior Backend Engineer (Elixir) - Remote

This document contains tailored responses for the written questions in the Remote job application.

1. What makes you interested in working with Remote?

I am deeply drawn to Remote for two primary reasons: its mission of democratizing global talent and its pioneering, forward-thinking approach to engineering delivery.

First, as someone who has worked in distributed, multicultural teams (such as Citibank and Perficient/Caterpillar), I have experienced firsthand the incredible value of global collaboration. Remote is solving one of modern business's most significant challenges—removing geographical barriers so that companies can recruit and manage teams compliantly, bringing life-changing opportunities and local wealth to talent worldwide.

Second, I am highly inspired by Remote's explicit commitment to innovation and AI. Your mandate to *"redesign how engineering work ships with autonomous agents as the default execution layer"* perfectly mirrors my own software engineering philosophy. Over the past few years, I have actively operated and customized agentic harnesses (such as Pi.dev and opencode) in my daily workflow to automate repetitive coding tasks while applying rigorous human architectural oversight, testing, and systemic verification. Working at a company that is actively building these agentic workflows into the core engineering fabric is exactly where I want to focus my career.

2. Remote is a value-driven company. Have you seen our values page? (<https://remote.com/notion/d43475e8895c4b7c811490aef64782b3>) What resonates with you most?

Of Remote's core values, **Transparency** and **Excellence** resonate with me most deeply.

- **Transparency:** As an asynchronous-first team, transparency is not optional; it is the absolute foundation of trust. Keeping all technical decisions, documentation, goals, and conversations open and accessible ensures that every team member remains aligned, empowered to take action, and fully capable of operating independently and synchronously toward collective goals.
- **Excellence:** I firmly believe that true excellence is a collaborative sport. I am a strong advocate for teamwork because a single mind working in isolation is nothing compared to the power of multiple minds thinking and building together. True excellence arises when diverse perspectives collaborate, brainstorm, and challenge one another through robust design discussions and constructive code reviews, elevating the quality of the final product far beyond what any single engineer could achieve alone.

The synergy of these values creates an incredible work-life balance, allowing engineers to put their family and life first while delivering world-class outcomes asynchronously.

3. Our backend is built in Elixir. Could you share details about your experience with the language and your thoughts on it? If you don't have experience with Elixir, how would you approach learning it, and what do you find appealing about functional programming?

I have hands-on, professional experience developing and architecting with **Elixir and Phoenix Framework** in my recent role as a Lead Software Engineer for a Mobility Startup. To continuously expand my modern Elixir toolkit, I am also currently reading a book on the Ash Framework and exploring its ecosystem.

My experience with Elixir has made it one of my favorite backend languages. I find several of its core paradigms incredibly compelling: * **Concurrency and Fault Tolerance:** The Erlang VM (BEAM), lightweight processes, and OTP supervisor trees make building self-healing, highly concurrent, and fault-tolerant applications incredibly simple compared to traditional thread-based ecosystems. * **Functional Purity & Predictability:** The combination of immutability, pattern matching, and the pipe operator (`|>`) leads to codebases that are clean, highly readable, and exceptionally easy to test and debug. * **Rapid Productivity:** Phoenix enables rapid full-stack iteration without

sacrificing the performance, low latency, and low memory consumption necessary for high-throughput production environments.

In my mobility startup role, I designed and maintained Elixir-based backends that integrated complex third-party APIs and managed persistent database transactions with PostgreSQL, demonstrating how Elixir can power critical infrastructure smoothly.

4. Can you share an example of a product or project you led through collaboration with stakeholders, product, and design? Tell us about it.

In my recent role as a **Tech Lead & Lead Software Engineer** at a high-growth **Mobility Startup**, I led the end-to-end architecture and launch of our core on-demand transportation platform.

- **The Challenge:** Building a real-time, highly reliable mobility application (mobile apps for passengers/drivers and a web-based back office) required resolving complex logistical and technical puzzles. However, the greatest challenge lay in bridging technical execution with commercial viability, real-time dispatch operations, and strict regulatory compliance.
- **Collaboration & Continuous Communication:** To align our engineering roadmap, I established a system of continuous communication and collaboration with key stakeholders across departments:
 - **Commercial Engineer & General Manager (Sales Expert):** I collaborated daily to understand customer acquisition strategies and monetization goals. I translated pricing tiers, commission models, and sales performance indicators into concrete technical specifications and backend calculations.
 - **Operations Manager:** We worked together to design real-time dispatcher dashboards, mapping integrations, and fleet monitoring tools, ensuring operational realities matched the software's performance and driver dispatch algorithms.
 - **Legal Department:** Mobility is highly regulated. I worked closely with our legal team to map out data protection regulations, passenger liability terms, and electronic contract signature flows, ensuring our database schemas and API security designs were fully compliant before the public launch.
 - **Product & Design:** I translated these multi-department inputs into actionable developer specs, partnering with UX/UI designers to craft frictionless driver and rider journeys (built on Flutter) and guiding our development team to implement them successfully.
- **The Outcome:** Thanks to this continuous, cross-functional collaboration, we successfully launched a fully compliant, highly scalable mobility platform on schedule. By aligning technical design with business, operational, and legal guidelines, we achieved excellent user retention, optimal dispatch efficiency, and a solid foundation for continuous feature expansion.

5. Tell us about your proudest achievement from your last two years of work. Why are you proud of it?

My proudest achievement from my last two years of work was **architecting, building from scratch, and successfully launching into active production a complete, end-to-end carpooling ecosystem**—available live at <https://app.tellevoapp.cl>.

- **The Scope and Architecture:**
 - **Spring Boot Java Backend Services:** I designed and developed a robust, secure, and transactional backend from scratch using Java and Spring Boot. This backend handles critical operations including real-time ride matching algorithms, passenger/driver coordination, user authentication, and secure transaction tracking.
 - **DevOps & Cloud Deployments (VPS & Google Cloud Run):** To ensure a cost-efficient yet highly scalable deployment strategy, I deployed the core application components across high-performance **VPS** environments, and successfully migrated and configured services on **Google Cloud Run** to leverage serverless auto-scaling and high availability.
 - **2 Mobile Flutter Applications:** I engineered and published two distinct, feature-rich mobile applications using Flutter—one dedicated and optimized for drivers to manage trip plans and passenger dispatches, and another for riders to book and track trips in real time.

- **Complete SPA / PWA Frontend:** Finally, to maximize user accessibility across all device types and eliminate app-store friction, I built and launched a fully featured web version of the product as a Single Page Application (SPA) and Progressive Web App (PWA), which lives at <https://app.tellevoapp.cl>.
- **Why I am proud of it:** I am exceptionally proud of this project because it represents the ultimate full-cycle engineering achievement. Taking a highly complex mobility idea and single-handedly architecting the system, writing the code for two mobile apps, building a comprehensive web PWA, developing the backend services in Java, and managing the multi-environment deployments to VPS and Google Cloud Run was an incredible journey. Seeing the platform successfully launched in production with active users relying on it daily has been the most rewarding validation of my technical execution and architectural judgment.